Mohammad Nusairat

224-517-5265 | mnusairat2003@gmail.com | linkedin.com/mohammad-nusairat | mohammadnusairat.com | Rolling Meadows, IL

EDUCATION

University of Illinois at Chicago **Chicago, IL**
B.S. in Data Science with a Concentration in Computer Science, GPA: 3.50 *May 2025*

- Related Coursework: Data Structures & Algorithms, Advanced Data Structure Practicum, Intro to Machine Learning, Software Design, Database Systems, Optimization for Analytics, Business Project Management

University of Illinois Urbana-Champaign **Champaign, IL**
M.S. in Computer Science, GPA: 4.0 *May 2028*

- Related Coursework: Applied Machine Learning

SKILLS

Programming Languages: Python, Javascript, TypeScript, C++, C, C#, Java, SQL, R, SAS

Technologies: Flask, React, Next.js, Node.js, Tailwind CSS, Railway, Git, MongoDB, PyTorch, TensorFlow

EXPERIENCE

Constellation Energy Corporation **Remote**
Jr. Full-Stack Developer *August 2025 - Present*

- Building a Modern AI-Powered Sales Platform.

MUHSEN **Remote**
Software Engineering Intern *June 2025 - Present*

- Built and deployed an **internally adopted org-wide** CSV processing platform for donor data encryption, and **advanced data cleaning** and **deduplication**.
- Improved data privacy compliance and volunteer workflow by 10x, by implementing **column-level encryption** using Fernet and **domain-gated Flask endpoints**.
- Reduced duplicate donor records by 40% through **fuzzy matching** and **entity resolution**, using recordlinkage and rapidfuzz for **clustering** and ID assignment.
- Increased donor outreach accuracy by **validating 5,000+ emails and addresses**, by integrating ZeroBounce API and Google Places API into a custom Flask + Next.js suite.

Dasion **Remote**
Machine Learning Intern *May 2025 - August 2025*

- Built a **multimodal cancer imaging platform** used to manage patient records and analyze MRI, CT, X-ray, and histopathology scans, by developing a full-stack app with FastAPI, PostgreSQL, and React.
- Enabled **4-way breast cancer classification with 91% confidence**, integrating a **CNN-based histopathology model** into the platform for real-time tissue analysis.
- Achieved **68% micro-F1 score** on unseen protein networks, by training a **GraphSAGE model** with PyTorch Geometric for **multi-label PPI classification across 24 disjoint graphs**.
- Enhanced model interpretability and diagnostic accuracy by generating **t-SNE visualizations** and **error heatmaps** to **evaluate learned protein embeddings**.

MailSped **Remote**
Back-End Software Engineering Intern *June 2025 - July 2025*

- Expanded the AI-powered email platform, integrating Outlook via **Microsoft Graph API** and **Azure OAuth2**
- Performed maintenance by **resolving critical bugs**, took part in **Agile Sprints** and **Weekly Scrum** Meetings
- Implemented **TTL indexing** in **MongoDB** for efficient auto-expiration, streamlining stored email cleanup

Comprehension, Collaboration and Creativity Lab **University of Illinois at Chicago, IL**
Undergraduate Research Assistant *Sept 2024 - May 2025*

- Research cognitive patterns in **algorithmic** problem-solving for a study on “Aha! moments” in programming
- Analyze participant interviews, **annotated** insight-related data, **co-authoring** findings **presented** at MPA 2025

Outlier **Remote**
Software Developer - AI Trainer *July 2024 - April 2025*

- Evaluate the quality of AI-generated Python/C++ code by developing **robust test cases** to train the model
- Enhance the accuracy of LLM’s code through **data ingestion** by solving complex algorithmic problems

Headstarter AI **Remote**

Software Engineering Fellow *July 2024 - August 2024*

- Built 5 **AI apps** & APIs using NextJs, OpenAI, Pinecone, StripeAPI with 98% accuracy as seen by 1000 users
- Developed projects from design to deployment **leading 4 engineering fellows** using **MVC** design patterns
- Coached by Amazon, Bloomberg, and Capital One engineers on **Agile**, **CI/CD**, **Git** and **microservice** patterns

PROJECTS

Arabic YouTube Transcript Translator *Website*

- Built a **React + Vite** frontend and **Dockerized Node.js + Express.js** backend web app that transcribes Arabic YouTube videos, translates them into English, and generates timestamped PDF, Markdown, and SRT files
- Streamlined content creation for my personal business by **reducing Arabic content preparation time by 86.7%** and **increasing transcription and translation accuracy by 20%** as tracked by internal QA logs
- Achieved **50k+ LinkedIn views** with **330+ users** and **16+ hours transcribed** on launch day

Campus Meeting Point Application *Website*

- Engineered a full-stack web app **significantly reducing** meeting problems on UIC's undergraduate campus
- Built using **Flask** and **React**, the app uses **XML parsing of OpenStreetMap data** to calculate the optimal meeting point between any number of users on campus, filtered by user-selected building utilities and features
- Developed scalable **REST API** endpoints supporting building autocomplete, tag-based filtering, and geospatial pathfinding using **Dijkstra's algorithm** and **Geodesic Fermat point logic**; deployed via Railway and Vercel
- Engineered fault-tolerant location logic and dynamic frontend components for search, filters, and ETA summaries, enabling real-time user coordination with location-aware filtering via **NetworkX** and **geopy**

Book Review Application *Website*

- Built a full-stack web app with **user auth**, **search**, **review features**, and Railway MySQL **cloud deployment**
- Integrated **advanced SQL logic** like search filters and average rating via GROUP BY, LIKE, ORDER BY
- Seeded **100+ book records** using Python + Pandas; enabled dynamic CSV-to-SQL loading with error handling
- Built secure login/registration with **SHA-256 hashing** and session-based access control
- Implemented **dynamic search** + review UX with Flask, Jinja templates, and modular route handling

PUBLICATIONS

Rahman, J., Virruso, A., Nusairat, M., Khan, R., Miller, T. S., & Wiley, J. (2025, April 10-12). Examining the Aha! experience in computer programming. Poster presented at the 97th Annual Meeting of the Midwestern Psychological Association, Chicago, IL